SoI Pre-Demo Day Pitch Deck

(Pre-Incubation | Current Stage: Proof of Concept)

Project Title: Blockchain Academic Credential System
Team: Block Master’s | SoI Lab Code: BW |Cohort: 2024-28
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Pre-Demo Day: Jan 6–7, 2026

Tip: Keep this deck to ~10–12 slides for the live pitch. Put extra details in Appendix slides.

Pitch in One Sentence + SoI Process (Where We Are)

Pitch (one sentence): "We help academic institutions and students achieve secure, instant, a tamper-proof credential verification by leveraging blockchain technology for digital signatu and biometric authentication, reducing fraud and administrative burder

Problem & End-User	Ideation	PoC (Current)	Prototype (Next)	MVP	Final Solution

Show the jury what you have completed so far (Problem → Ideation → PoC) and what comes next (Prototype → MVP).

Problem & End-User
(What we validated)

- End-user: Students, Academic Institutions, Employers
- Context: Global academic landscape, post-graduation job market, need for trustworthy credential validation.
- Evidence: High rates of credential fraud, slow manual verification processes, administrative overhead.
- Success metric: Reduction in verification time, elimination of credential fraud.

Ideation
(What we explored)

- Ideas explored: Centralized databases, advanced watermarking, blockchain-based solutions.
- Evaluation: Blockchain offered superior security, transparency, and decentralization.
- Why we chose this idea: Addresses core issues of trust and immutability in credential verification.

PoC
(What we proved)

- PoC scope: Digital signatures for students & institutions, wallet creation, document hashing, verification, and biometric-secured private keys.
- Key metric results: 100% hash matching accuracy, signature creation <5s, verification <2s.
- Demo link/QR: [ ]
- Key learning:Biometric access improves security and user experience without affecting decentralization.

Problem & End-User (Validated Pain)

Problem (≤120 words)

“Today, the academic credential verification process is plagued by fraudulent certificates, lengthy manual checks, and high administrative costs. Institutions struggle to maintain the integrity of their issued degrees, while students face delays in getting their qualifications recognized. Employers, too, spend significant resources verifying credentials, often with unreliable results. This lack of trust and efficiency undermines the value of education and creates barriers for students seeking employment.“

What happens today?:Manual verification, paper certificates, high forgery risk, and slow processing.

Why does it matter (impact)?:Loss of trust in degrees, financial loss from fraud, delays in student careers, and heavy workload for institutions.

How often / how severe?:A global and widespread problem, affecting millions of academic credentials every year.

End-User + Context

Target user/customer:Universities colleges, vocational schools, students employers

Where/when:Credential issuance, job application process, academic transfers.

Trigger: Need for secure and verifiable proof of academic achievement

Current workaround: Manual checks, third-party verification services (often slow and costly).

Evidence & Success Metrics

Evidence:(e.g., "# interviews with registrars highlighting fraud concerns," "# observations of manual verification delays," "reports on the rising cost of credential fraud," "quote from a university official: 'We need a bulletproof way to verify degrees swiftly.'")

User-level success metric(s): [e.g., 3-5 days for verification, 15% fraud rate] → Target (with your solution): [e.g., <1 min for verification, 0% fraud rate]

Research & Existing Solutions (Literature + Market)

Literature / Prior Research

• Existing digital credential systems often rely on centralized databases, creating single points of failure.
•Blockchain technology offers inherent immutability and transparency, ideal for secure record-keeping.
•Biometric authentication is a robust method for secure access to digital assets.
Sources: Cite relevant academic papers, industry reports on blockchain in education, biometric security standards

Existing Products / Services

• Solution A: Traditional paper certificates / manual verification — limitation: Prone to fraud, slow, high administrative cost.
• Solution B: Centralized digital credential platforms — limitation: Single point of failure, data privacy concerns, potential for manipulation.
• Solution C: Basic blockchain credential systems (without advanced signing/biometrics) — limitation: May lack robust user-friendly signature processes and advanced private key management.

Gap / Opportunity + Why Now

Gap we found: "While blockchain is gaining traction, a comprehensive solution integrating user-friendly digital signatures with secure biometric private key access for academic credentials is still missing."
Our USP (unique selling proposition): "Our system provides an unparalleled level of security and user convenience by combining blockchain's immutability with a seamless digital signature process secured by biometric private key access."
Why now:"Increasing digital transformation in education, heightened awareness of cybersecurity, decreasing costs of blockchain infrastructure, and growing demand for verifiable digital identities make this the ideal time."

Ideation Snapshot (3+ Ideas) + Selection Rationale

Show at least 3 distinct solution ideas. Explain why the selected idea wins on desirability, feasibility, and viability.

Idea	Desirable?	Feasible ?	Viable?	Notes / Why or Why Not
Idea 1: Centralized Digital Vault	[H]	[H]	[M]	Easy to implement but lacks true security and trust.
Idea 2: Basic Blockchain Credentialing	[H]	[H]	[H]	Strong security, but the private key management and signing process could be cumbersome for end-users.
Idea 3: Blockchain + Biometric Signature	[H]	[H]	[H]	Secure, user-friendly, and scalable.
Selected: Blockchain + Biometric Signature	[H]	[H]	[H]	Offers highest security with biometric ease, practical to build, and suitable for real-world adoption.

Selected Idea: Blockchain Academic Credential System with Biometric Signature
1-2 sentence justification: "This idea was selected because it provides the strongest user value through enhanced security and ease of use (biometric access), is feasible within our lab resources (blockchain development + biometric integration APIs), and has a clear path to payer (institutions seeking robust verification solutions)."
PoC focus (what we decided to prove first): "The core assumption we proved was the ability to securely generate wallets, digitally sign credentials using institution's public key, verify signatures through hash matching, and enable private key access via biometric authentication."

Evolved Solution + Value Proposition

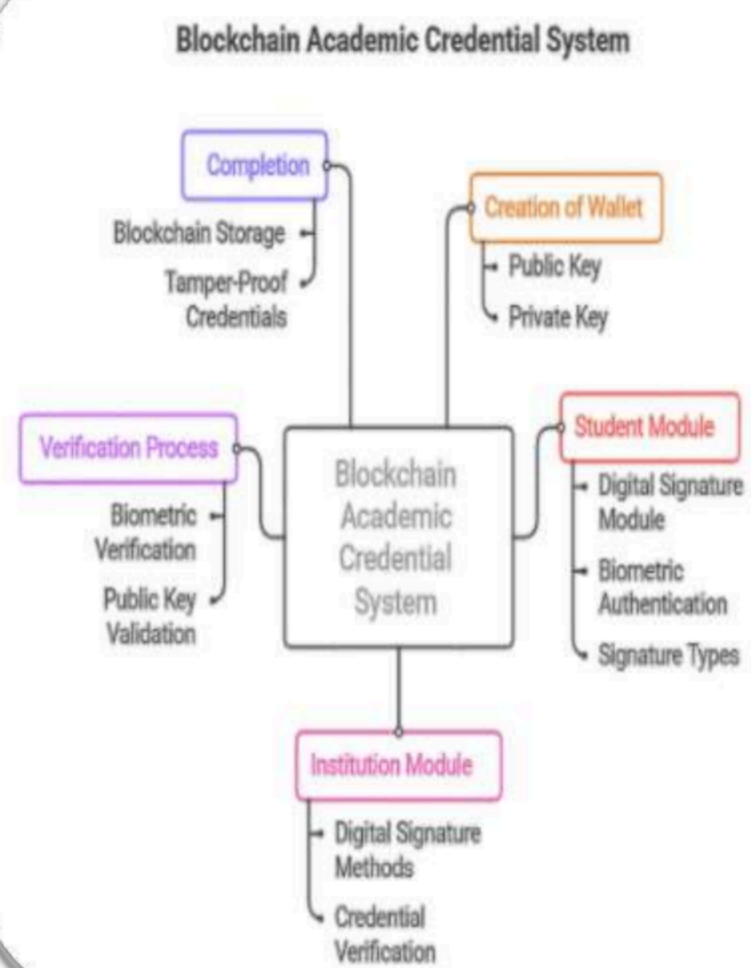
What it is

• Institutions issue tamper-proof blockchain credentials with digital signatures and hash verification, while students access and share them using biometrics.
• What changed since ideation: Added biometric private key access and standardized unstructured data into verifiable hashes.
• Key differentiators:
  - Blockchain-based digital signatures
  - Biometric-secured private key access
  - Fast, hash-based credential verification

Value Proposition

For [academic institutions, students, and employers], we enable [secure, instant, and tamper-proof credential verification] by [leveraging blockchain for digital signatures, unique hash-based verification, and biometric authentication for private key access].Compared to [current alternatives like manual verification or centralized digital systems], we are better because:
• Eliminates credential fraud with immutable blockchain records
• Provides unparalleled user convenience through biometric private key access.
• Provides unparalleled user convenience through biometric private key access.

How it Works (Diagram / UI / Flow)

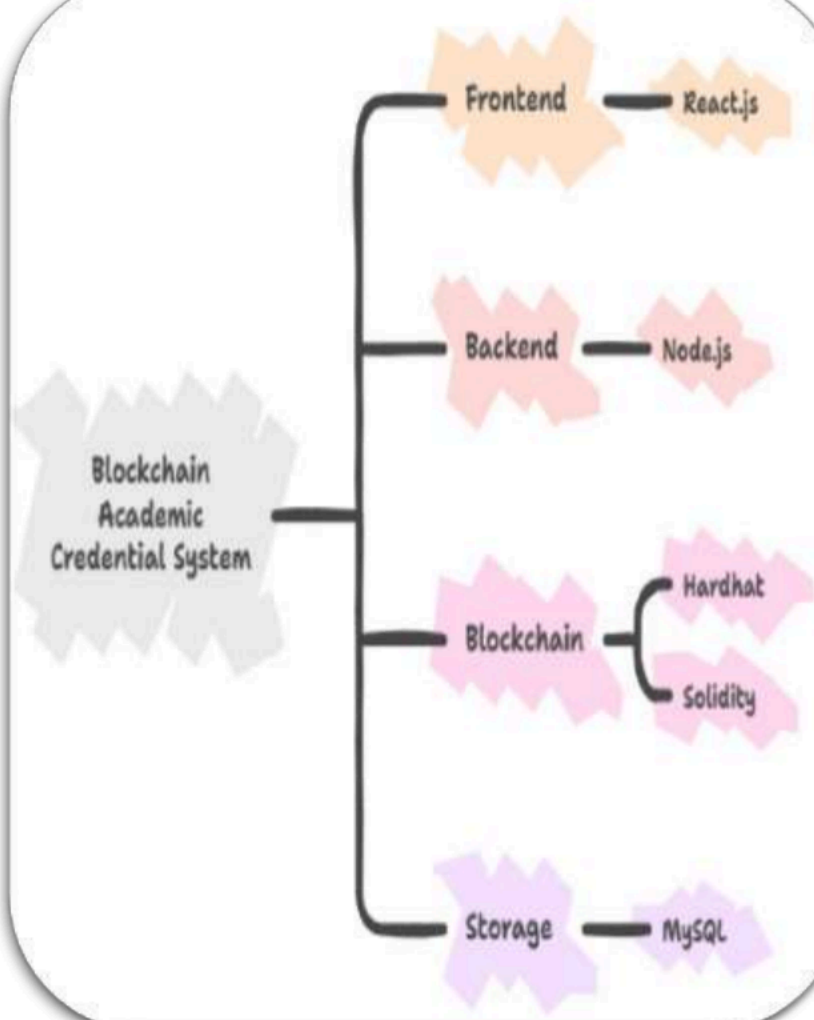


PoC Scope & What We Built (Current Stage)

PoC Scope (In / Out)

In scope: User (student/institution) account creation with public/private wallet generation. Institution issuing credentials signed with its public key. Blockchain transaction for credential issuance. Biometric (fingerprint/face) authentication for private key access. Conversion of unstructured credential data to a verifiable hash. Hash comparison logic for digital signature verification.
Out of scope (for now):Full UI for all user roles. Integration with existing student information systems (SIS).Scalability testing for millions of transactions. Regulatory compliance frameworks.
Assumptions/constraints: Using a public blockchain (e.g., Ethereum testnet) for initial PoC; focus on core security and verification logic.

PoC System Overview



Prioritized Features (for PoC)

Must-have (built):
- Wallet generation for users.
- Digital signing of credentials by institution.
- Blockchain record of issuance.
- Biometric unlock for private keys.
- Hash-based credential verification.

Should/Could (deferred):
- Batch credential issuance.
- Revocation mechanism.
- Multiple credential types.

PoC Evidence (Results) + Demo Link

Key Metrics (targets vs. observed)

Critical assumption / test	Metric	Target	Observed (PoC) + evidence link
Credential signing speed	Time from input to blockchain record	<10s	<5s
Verification accuracy	Hash match success rate	100% hash match	100%
Biometric private key access	Success rate / Latency	99% success / <2s latency	98% / <1s
Fraud prevention	Incidence of false positives	0% false positives	0% in controlled PoC [N/A]

Key Findings / Learnings

- What worked: What Blockchain issuance, digital signatures, and biometric access worked securely and reliably. Hash conversion and verification were fully accurate.
- What did not work (yet): Integration with existing academic record systems (SIS) was complex and out of scope.
- What changed because of this learning: Future development will include standardized APIs/middleware for easier SIS integration.
- Biggest open risk: User adoption by institutions and navigating regulatory requirements.

Demo Video + Repo

Demo video (≤2 min): [URL]

QR: [paste QR image]

Code / CAD / notebook repo (read-only): [URL]
Data/logs (if any): [URL]

Tip: Include timestamps in the video for key moments.

Viability: Who Pays + Go-To-Market

Who Pays & Why

Customer segment: Academic institutions (universities, colleges, certification bodies), potentially employers for bulk verification services.
Buyer / payer: University registrars, IT departments, administrative heads
Procurement path: Direct sales, partnerships with education technology providers, government tenders for educational digitization.
Pricing hypothesis:
• / month] subscription for institutions based on student/credential volume. OR
•[Y / unit] hardware (e.g., secure biometric device, if needed) + [Z / month] service fee for enterprise clients.
Value to payer (ROI):
• Significant reduction in fraud detection costs, faster verification (time saved), enhanced institutional reputation (safety/trust), potential for compliance with future digital identity regulations.
• Reduction in verification cycle time, decrease in reported fraud cases, percentage of credentials issued digitally.

Go-To-Market (Initial Plan)

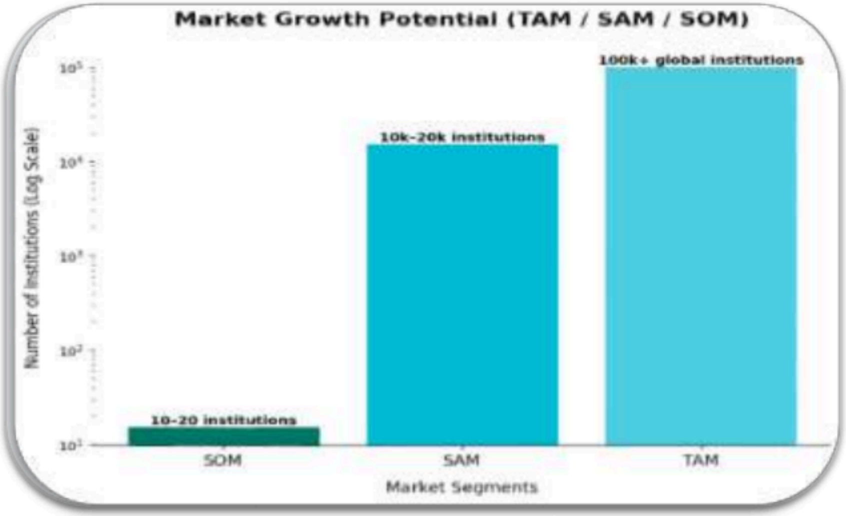
• Beachhead (your first small target group): Pilot program with 1-2 local universities/colleges (e.g., our campus / a specific department).
• Channel: Direct outreach to university IT and registrar offices, participation in education technology conferences, partnerships with existing student information system (SIS) providers.
• Key stakeholders: University CIOs, Registrars, Admissions Officers, Government education departments.
• Adoption plan: Pilot setup with selected institutions, define KPIs for success (e.g., 90% digital credential issuance within pilot, 50% reduction in verification time), establish a feedback loop for continuous improvement.
• Regulatory/ethics/safety notes (if any): Adherence to data privacy regulations (e.g., GDPR, FERPA) for biometric and personal data. Clear consent mechanisms for biometric data collection and usage. The system itself enhances safety against fraud.

Growth Potential (Market + Scalability)

Market (TAM/SAM/SOM) — rough estimates ok

TAM: Global academic sector issuing credentials (billions of credentials annually, hundreds of thousands of institutions).
SAM: Institutions willing to adopt blockchain for credentials (e.g., 10-20% of global institutions in 5 years).
SOM (12–24 mo): 1-2 major institutions in our region, 5-10 smaller institutions (e.g., 0.1-0.5% of SAM).
Assumptions: [e.g., # customers]: 10-20 institutions × [₹X/customer/year] (based on pricing hypothesis).

Visual (chart / funnel / map)



Growth & Scaling Plan

Growth levers (choose 3–5):
Expand to more sites/users within same customer (e.g., rolling out to all departments after a successful pilot in one).
Add adjacent use-cases/features (after PoC): e.g., skill badges, professional certifications, lifelong learning records.
Partnerships/integrations with existing education technology platforms and HR systems.
Lower cost / better performance through optimized blockchain solutions (e.g., private blockchain, layer 2 solutions).
New segments/geographies: Expand to corporate training, government certifications, or international markets.
Scalability considerations (tech/ops): "Migrating to a private blockchain for higher transaction throughput and lower fees, implementing sharding or layer-2 solutions for public blockchain scaling, optimizing biometric authentication infrastructure, and designing for modularity to support new credential types and integrations."

Roadmap + Ask (PoC → Prototype → MVP → Final

Roadmap (high level)

Now: PoC

- Proven: [metric]
- Demo ready
- Key learning: [ ]

Next: Prototype (Jan 2026)

- Build: [integration + enclosure/UI]
- Improve: [ ]
- Validate: [ ]

Then: MVP

- Small pilot with real users
- Measure KPIs
- Iterate

Later: Final Solution

- Production-ready build
- Deployment + support
- Scale

The Ask (what we want from the jury/advisors)

We are looking for

- Connections to academic institutions or organizations willing to participate in our Prototype pilot program.
- Expert advice on our private blockchain architecture and security protocols.
- Investment to develop the Prototype and scale the solution.
- Guidance on forming partnerships with EdTech companies or navigating institutional procurement processes.
- Mentorship on business model refinement and market strategy.

Success Criteria for the Next Stage (Prototype)

By end of Prototype stage we aim to achieve:

- A fully functional prototype demonstrating private blockchain integration, improved transaction speed (<1 second), and a comprehensive user interface.
- Positive feedback from pilot users on ease of credential issuance and verification.
- Initial legal and privacy review confirming compliance with relevant regulations for handling educational and biometric data.

Team & Roles (Why This Team Can Deliver)

Keep it simple: show role clarity + relevant skills + gaps and how you will fill them.

Name	Role	Key Skills / Contributions
Lead	Blockchain Architect	Smart contract development (Solidity), cryptography, system design, project management
Frontend Developer	UX	React/Vue.js UI/UX design, biometric integration (APIs).
Backend Developer	Integrations	Node.js/Python API development, database management, system integration.
Security Specialist	QA	Cybersecurity auditing penetration testing, quality assurance, blockchain security protocols.

Why Us (team advantage)

- Complementary skills: Blockchain, full-stack, security.
- Access to domain/users: Students and professors for feedback.
- Execution plan: Clear roles, agile development.
- Gaps + how we'll fill: Need legal and business support; will hire/mentor.

Mentor / Advisors

Mentor: Yuvedha Sri S
External advisors (if any): [ ]
Technical validators: [ ]

Contact: [email/phone]

Top Risks & Mitigation (Optional)

Focus on the biggest 3–5 risks that could stop you. Show a mitigation plan and what evidence you will gather next.

Risk	Impact	Likelihood	Mitigation / Next Test	Owner
Institutional Adoption Resistance	High	Medium	Run pilot programs, show ROI, 80% satisfaction.	Team Lead
Regulatory & Legal Hurdles	High	Medium	Get legal advice, make compliance checklist.	Security Specialist
Cybersecurity Vulnerabilities	High	Medium	Security audits, penetration testing.	Security Specialist
Scalability & Performance Issues	Medium	Medium	Optimize blockchain, test 100k transactions.	Blockchain Architect
Biometric Data Privacy Concerns	High	Medium	Encrypt data, get user consent, survey users.	Frontend Developer

If you deal with privacy/safety/ethics (AI, CCTV, biometrics, location tracking), state what data you collect and what you do NOT collect.